Machine Learning

Revision- Week 6

Objectives

- Revision

Where to find me?

Office:

Science Hall 250

Office Hours:

Weds and Thurs 11:00 AM - 1:00 PM EST or by appointment

Some Updates

- No office hours next Thursday, October 9th.

- To make up for it, I will be holding office hours on
  Monday, October 6th, 11:00 AM – 1:00 PM

Things to remember for Test 1 (Oct 7th)

- **Format of test questions is similar to quiz. We will also discuss some in class today**

- **Review lecture notes, lab.**

Exam Time: 9:30 AM – 10:45 AM

Location: SH 259

Things to remember for Test 1 (Oct 7th)

- **No electronic items** (cell phones, calculators etc.) are allowed.
- You are allowed to bring **3 pages** of notes
- **Cheating on an exam** will result in a score of **0**.

Exam Time: 9:30 AM – 10:45 AM

Location: SH 259

Things to remember for Test 1 (Oct 7th)

- Students **who are late** will not be granted additional time to complete the test.

- The **exam will not be rescheduled**. The only exception to this rule is a documented medical emergency, in which case you must provide a **medical note from a doctor or nurse practitioner** that clearly states **the date and time of your absence**.

Exam Time: 9:30 AM – 10:45 AM

Location: SH 259

Topics For Exam

- Different types of Data (Ordinal, Nominal etc).
- Supervised vs unsupervised
- Classification vs Regression
- Simple Probability Problems – Coin flipping, problems with dice
- Independent, mutually exclusive events, Venn Diagrams, conditional probability etc.
- Bayes Theorem.
- Naïve Bayes (in details), the assumptions for Naïve Bayes
- Linear Regression (with one variable/features vs multiple variables/features)
- Cost function for linear regression
- Gradient descent, learning rate, convexity
- Matrix formulation of linear regression
- Logistic regression (just sigmoid function and derivation from linear regression)

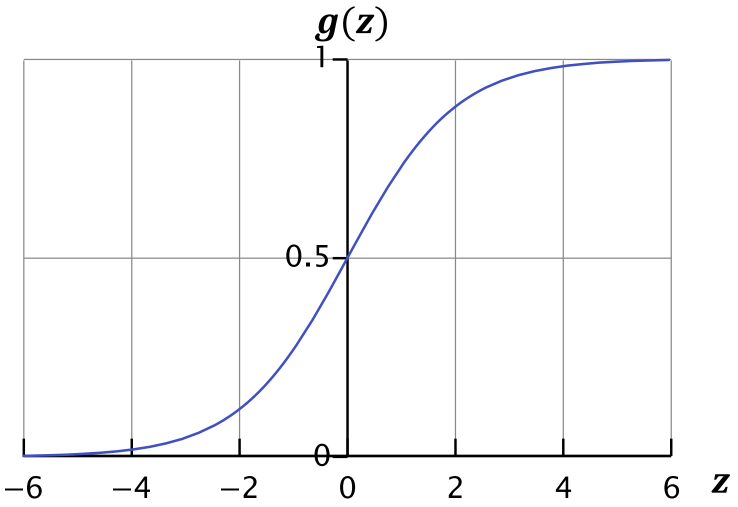
Logistic Regression

Want *0 <= ŷ <= 1,*
Linear regression equation : $f(x) = wT\ x + b$

Now we define logistic regression as $f(x) = g(wT\ x + b)$

Where $g(z) = \frac{1}{1+e^{-z}}$

$g(z)$ is called the **sigmoid** or **logistic** function



Questions

The core assumption of the Naive Bayes classifier is that all features are conditionally independent given the class label.

1. State the mathematical formula that captures the *likelihood* under this assumption.
2. Provide a real-world example from text or image data where this assumption is clearly violated.

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The core assumption of the Naive Bayes classifier is that all features are conditionally independent given the class label.

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1. $$P(y|x_1, x_2, \ldots, x_n) = \frac{\boxed{P(x_1, x_2, \ldots xn|y)} * P(y)}{P(x_1, x_2, \ldots xn)}$$

Likelihood

2. In a spam filter, the words "free" and "money" are highly correlated, especially in spam emails.

Questions

The Naive Bayes classifier is used to find the class that maximizes the posterior. Explain why the evidence term in the denominator is often ignored during the classification decision.

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The Naive Bayes classifier is used to find the class that maximizes the posterior. Explain why the evidence term in the denominator is often ignored during the classification decision.

$P(x_1, x_2, x_3, x_4, \ldots, x_n)$ is the **evidence** and is a constant for a given input across all classes . Since the classification decision depends only on the *relative magnitude* of the numerator (), dividing by the same constant is unnecessary

Questions

The Linear Regression cost function (MSE) involves squaring the prediction error.

1. Explain the primary consequence of this squaring operation regarding the model's sensitivity to a single, distant outlier, and
2. Suggest a simple alternative cost function (name the function) that would be more robust to such an outlier.

Questions

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1. The squaring operation places a very heavy penalty on large errors. This causes the regression line to be **highly sensitive** to single, distant outliers, pulling the line disproportionately towards them to minimize the squared residual.
2. Mean Absolute Error (MAE): $\frac{1}{n}\sum_{i=1}^{n} |\hat{y}_i - yi|$

Questions

You are given a large dataset of financial transaction records. The features include date, amount, and merchant category. Your task is to identify anomalous transactions that deviate significantly from typical spending patterns, but you have no pre-labeled examples of "fraud."

1. Is this problem best modeled as a **Classification, Regression, or Clustering** task
2. which of the three main ML paradigms (Supervised, Unsupervised, Reinforcement) does it fall under?

Questions

You are given a large dataset of financial transaction records. The features include date, amount, and merchant category. Your task is to identify anomalous transactions that deviate significantly from typical spending patterns, but you have no pre-labeled examples of "fraud." Is this problem best modeled as a **Classification, Regression, or Clustering** task, and which of the three main ML paradigms (Supervised, Unsupervised, Reinforcement) does it fall under?

1. Task Type: Clustering
2. ML Paradigm: Unsupervised Learning (since there are no pre-labeled examples of "fraud").

ANY QUESTIONS?